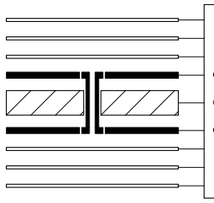
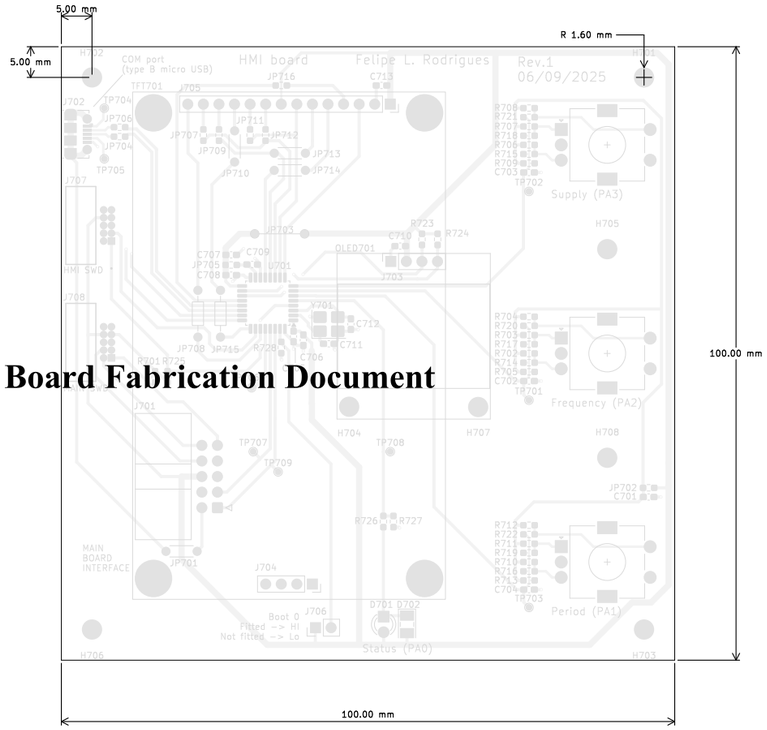
HMI Board Fabrication Document

Layer Stack Legend



Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
Core		1.48mm	FR4_7628	Dielectric	
Copper	**L2 (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.66mm
Note: external layer thicknesses are specified after plating



Top Fabrication (Scale 1:1)

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.

4) SURFACE FINISH: HAL LEAD-FREE

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLUE.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	100.000 × 100.000 mm
BOARD THICKNESS	1.660 mm
TRACE WIDTH	0.300 mm
TRACE TO TRACE	0.180 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	3.000 mm
ANNULAR RING	0.100 mm
COPPER TO HOLE	0.250 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.250 mm

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 5d038b1
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 1 of 12

HMI Board Fabrication Document

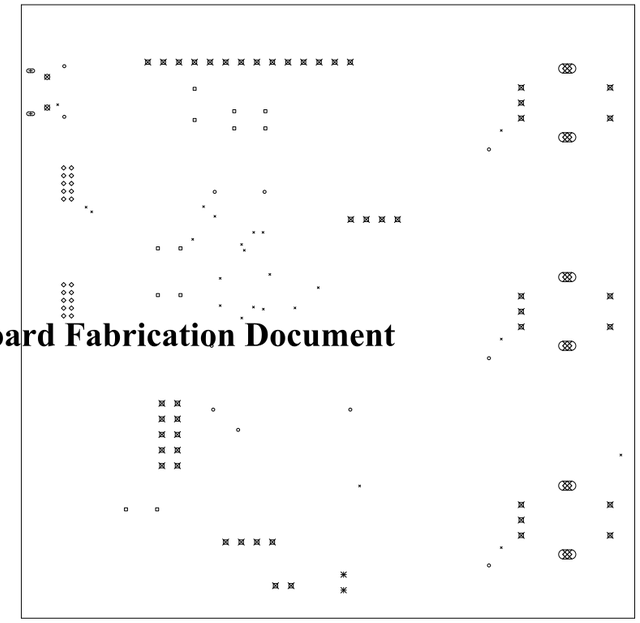
Bottom Fabrication (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 5d038b1
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 2 of 12

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	23	0.30mm (11.81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Via
○	11	0.50mm (19.69mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
+	2	0.60mm (23.62mils)	PTH	Slot	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
□	12	0.70mm (27.56mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
◇	20	0.74mm (29.13mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
⊠	2	0.85mm (33.46mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
✳	2	0.90mm (35.43mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
⌘	49	1.00mm (39.37mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
⌘	6	1.50mm (59.06mils)	PTH	Slot	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
	Total 127					



HMI Board Fabrication Document

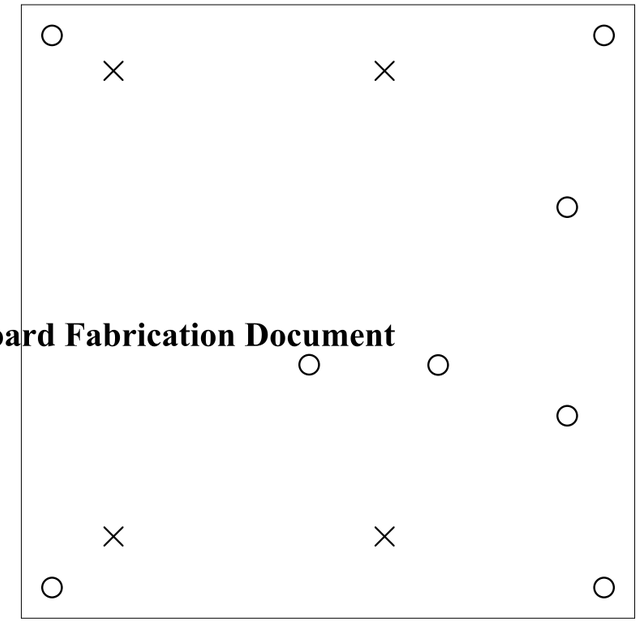
Drill Drawing L1 - L2 (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 5d038b1
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: Drill Drawing (L1 - L2)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 12

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	3.00mm (118.11mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Mechanical
○	8	3.20mm (125.98mils)	NPTH	Round	L1 (Slg, PWR) - L2 (Slg, PWR)	Mechanical
	Total 12					



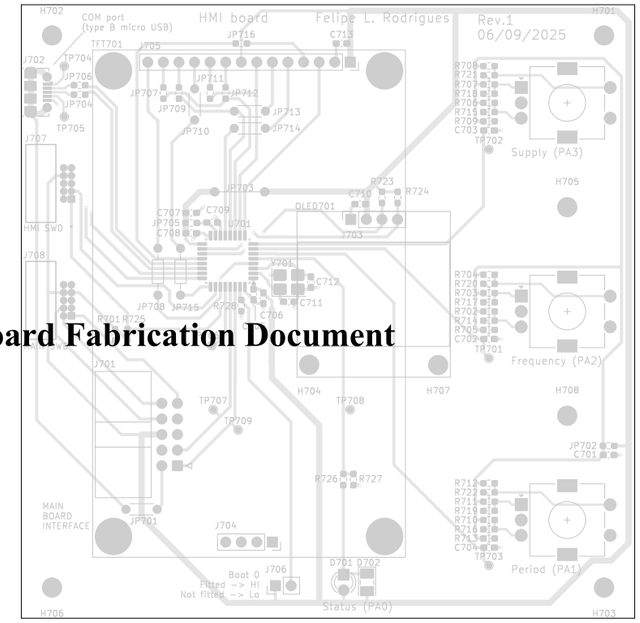
HMI Board Fabrication Document

Drill Drawing L1 - L2 (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 5d038b1
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: Drill Drawing (L1 - L2)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 4 of 12

HMI Board Fabrication Document



Ref.	Net	X [mm]	Y [mm]
TP701	FREQ	86.24	52.38
TP702	SUP	86.24	86.42
TP703	PER	86.24	18.60
TP704	Net-(J702-VBUS)	17.00	99.98
TP705	Net-(J702-ID)	17.00	91.73
TP706	Net-(U701-PB5)	41.03	54.41
TP707	LV_TX	41.28	44.00
TP708	ST_LED	63.63	44.00
TP709	LV_RX	45.34	40.70

Top Test Points (Scale 1:1)

Y
X

ChurrasCod Electronics

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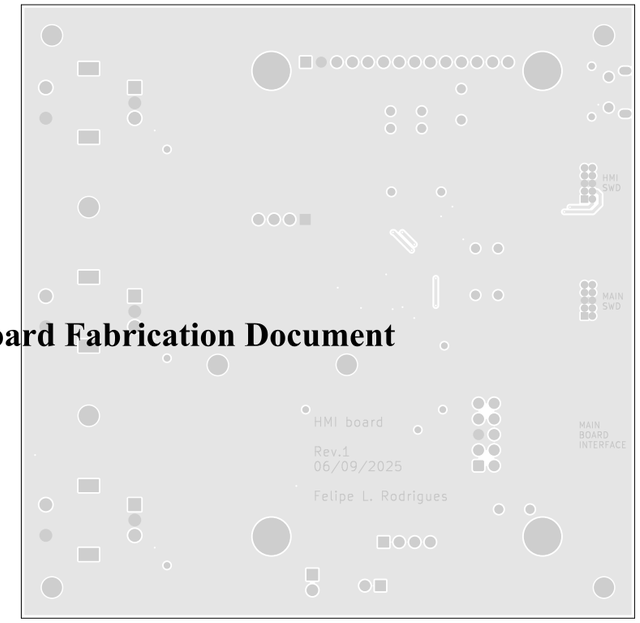
Sheet Title: Top Test Points (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 5 of 12

HMI Board Fabrication Document



Ref.	Net	X [mm]	Y [mm]

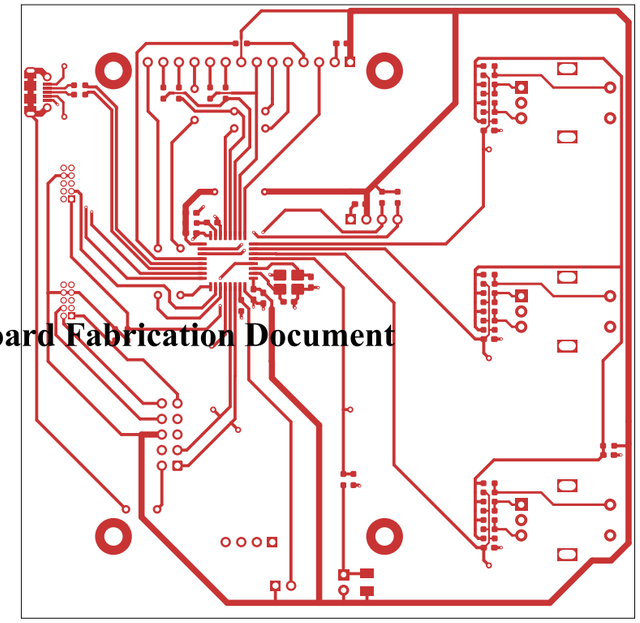
Bottom Test Points (Scale 1:1)

Y
X

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 5d038b1
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 6 of 12

HMI Board Fabrication Document

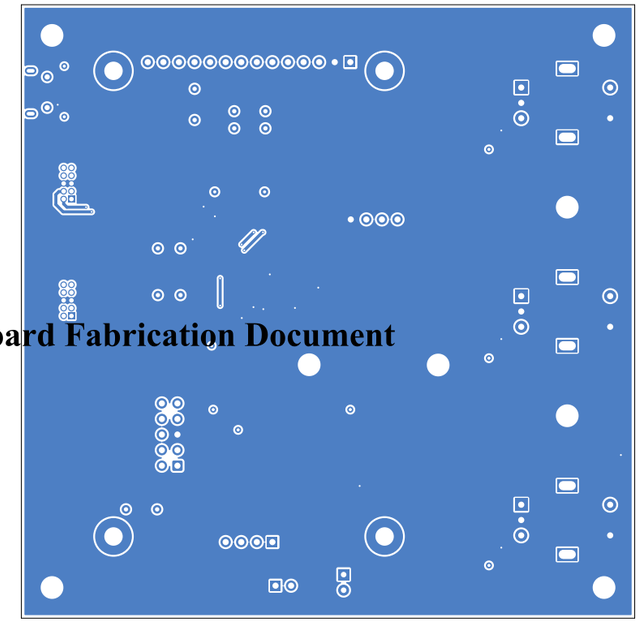


L1 (Sig, PWR) (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 5d038b1
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 7 of 12

HMI Board Fabrication Document

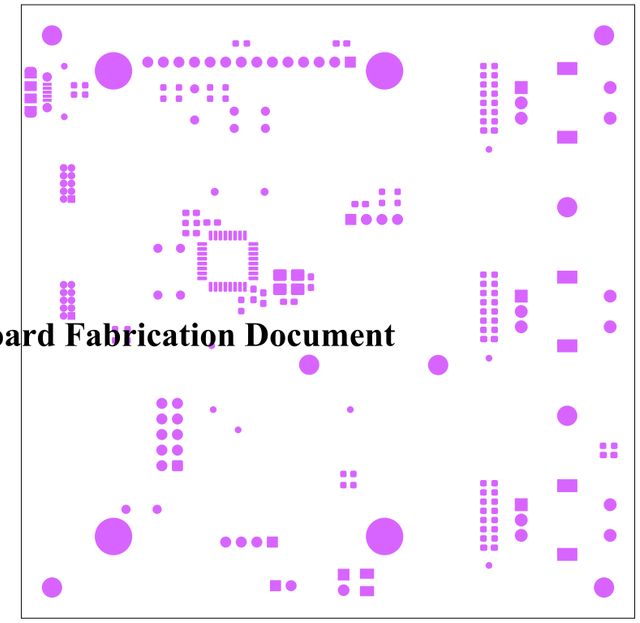


L2 (Sig, PWR) (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 5d038b1
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: L2 (Sig, PWR) (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 8 of 12

HMI Board Fabrication Document

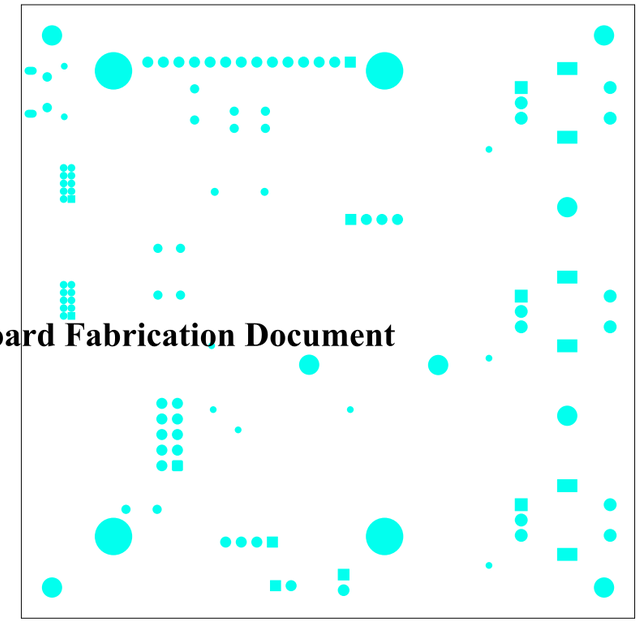


F.Mask (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 5d038b1
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: F.Mask (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 9 of 12

HMI Board Fabrication Document

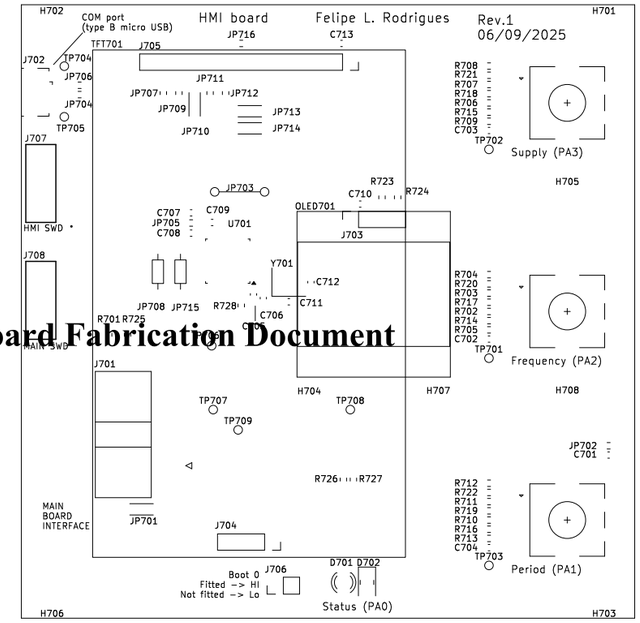


B.Mask (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 5d038b1
	Board Name: HMI Board		Project Name: 1 kV Strobe	
Sheet Title: B.Mask (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 10 of 12

HMI Board Fabrication Document

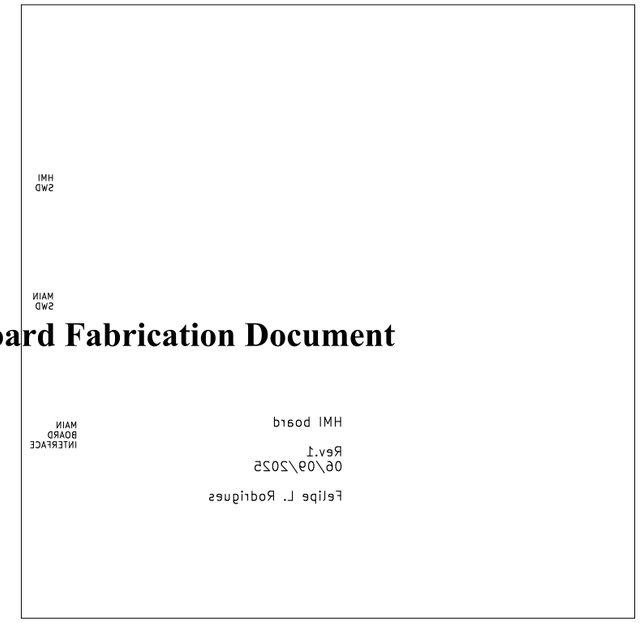


F.Silkscreen (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: **ChurrasCod Electronics**		Variant: PRELIMINARY	Git Hash: 5d038b1
	Board Name: **HMI Board**		Project Name: **1 kV Strobe**	
Sheet Title: F.Silkscreen (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **11** of **12**

HMI Board Fabrication Document



B.Silkscreen (Scale 1:1)

ChurrasCod Electronics

Comments:	Company: **ChurrasCod Electronics**		Variant: PRELIMINARY	Git Hash: 5d038b1
	Board Name: **HMI Board**		Project Name: **1 kV Strobe**	
Sheet Title: B.Silkscreen (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **12** of **12**